Single-cell & Spatial Transcriptomics in Cancer Research:

How Data Science Unlocks Biological Insight

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Lesson Materials

Github: https://github.com/khuongduying/ds_class_sc_spatial_20251129

References:

https://scanpy-tutorials.readthedocs.io/en/latest/pbmc3k.html

https://satijalab.org/seurat/articles/pbmc3k_tutorial

https://machinelearningcoban.com/2017/06/15/pca/

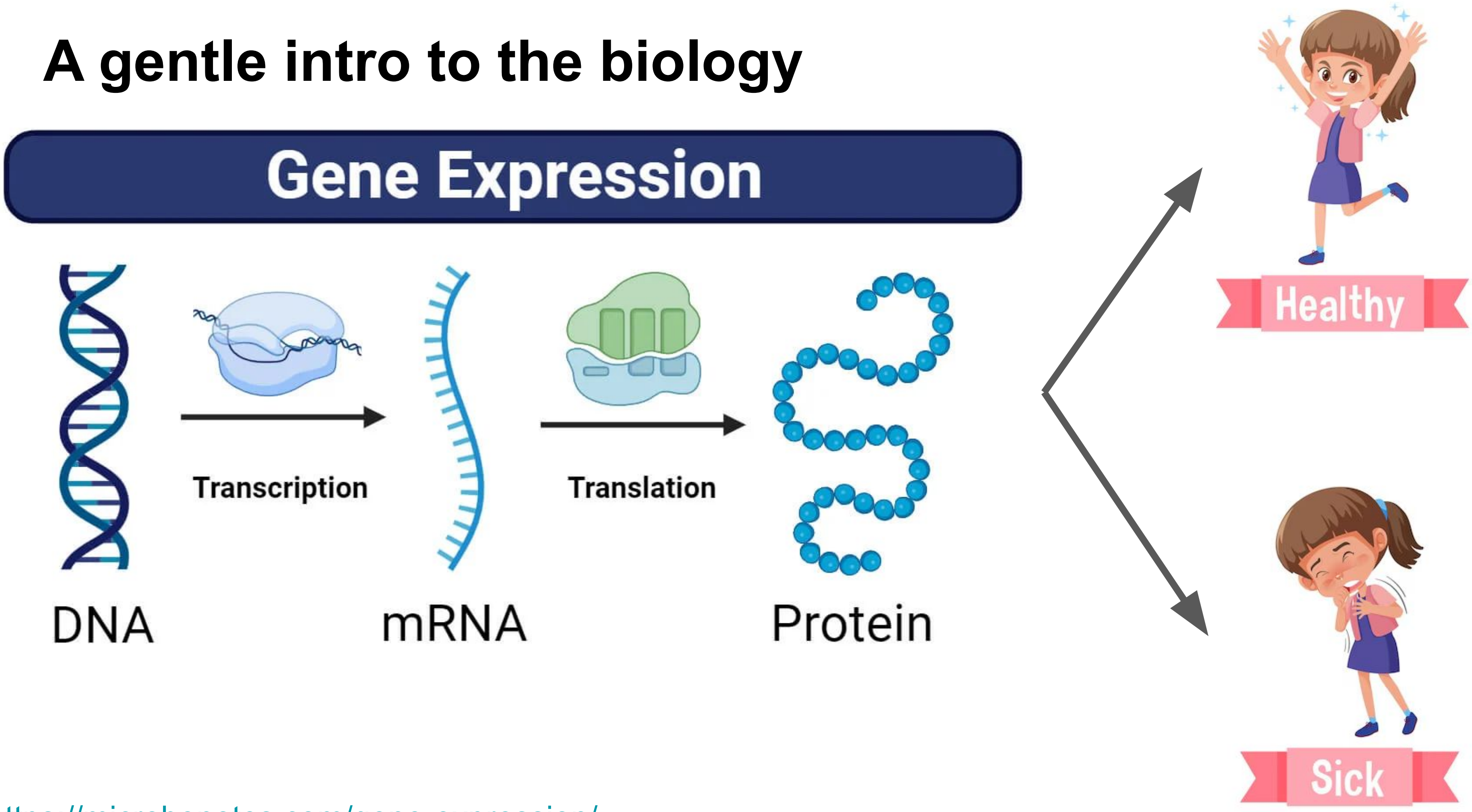
https://www.youtube.com/watch?v=aaANPo9fuuE

https://www.youtube.com/watch?v=OJU8GNW9grc

Content

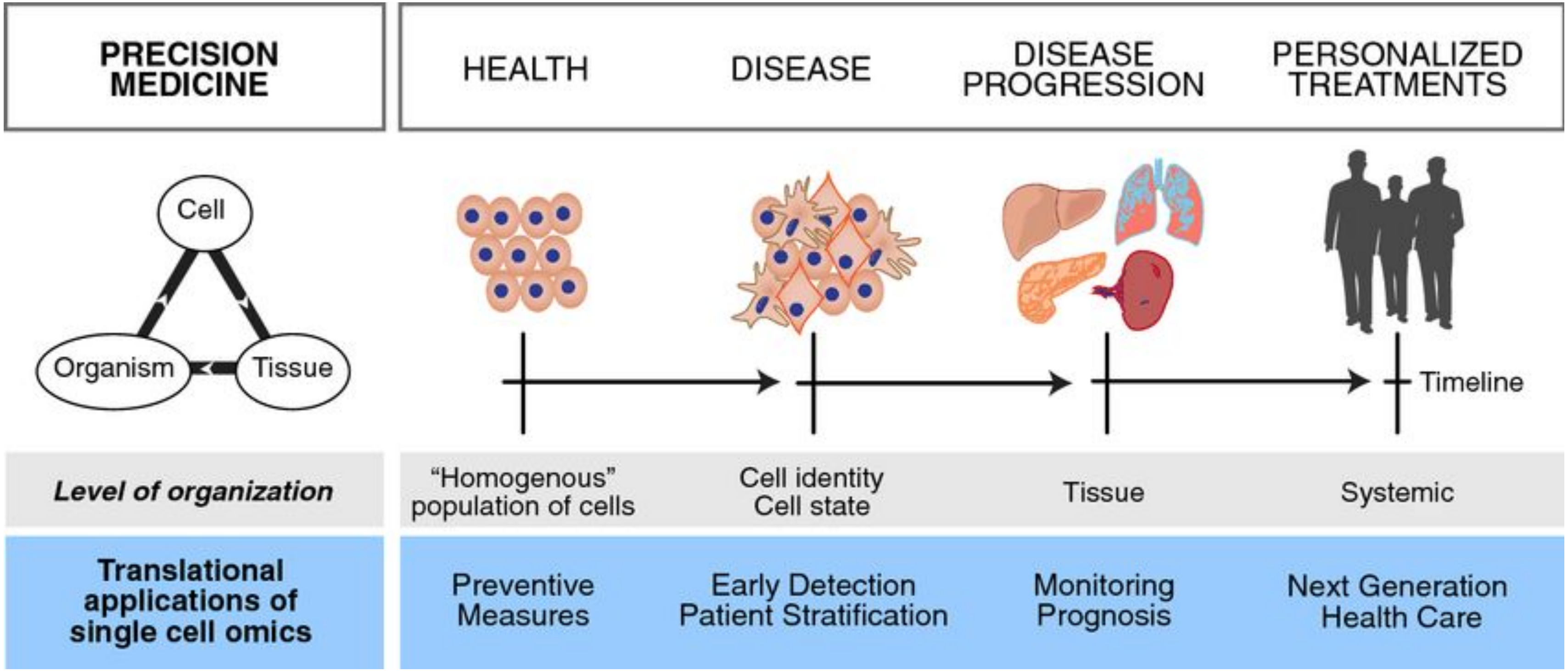
1. Unsupervised Learning with Single-cell Data
2. Spatial Autocorrelation in Spatial Transcriptomics

A gentle intro to the biology



https://microbenotes.com/gene-expression/

A gentle intro to the precision medicine



https://www.researchgate.net/figure/Single-cell-omics-is-advancing-the-development-of-personalized-treatments-Precision_fig1_340516110

Lung cancer & Tumor microenvironment (TME)

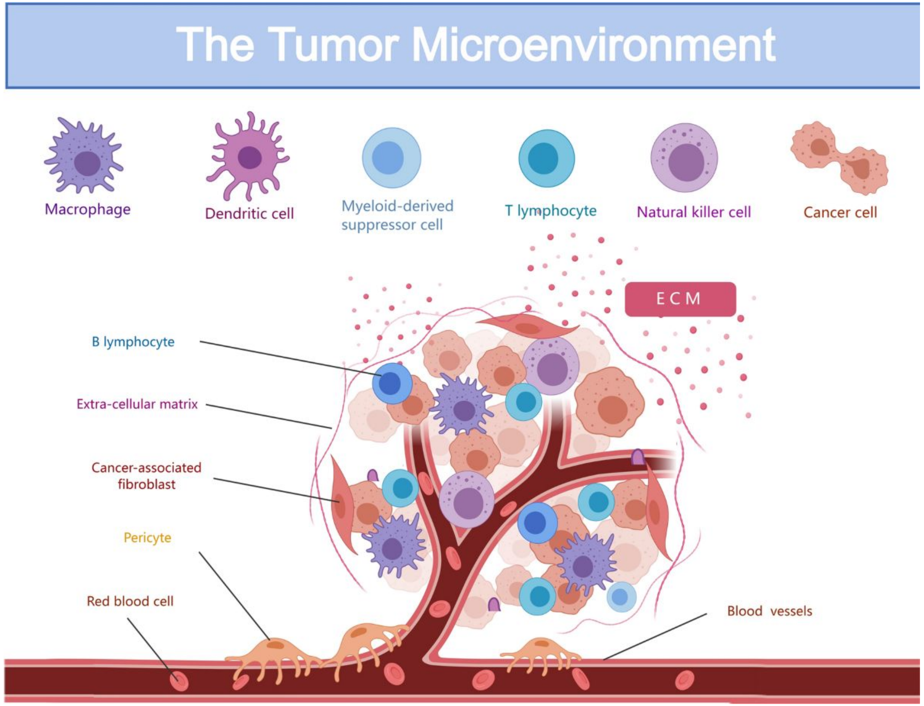
Non-Small Cell Lung Cancer (NSCLC)

- Most common type of lung cancer (~85% of cases) (WHO, 2023)
- Leading cause of cancer death worldwide (GLOBOCAN, 2022)

Tumor Microenvironment (TME)

Complex ecosystem surrounding tumor cells

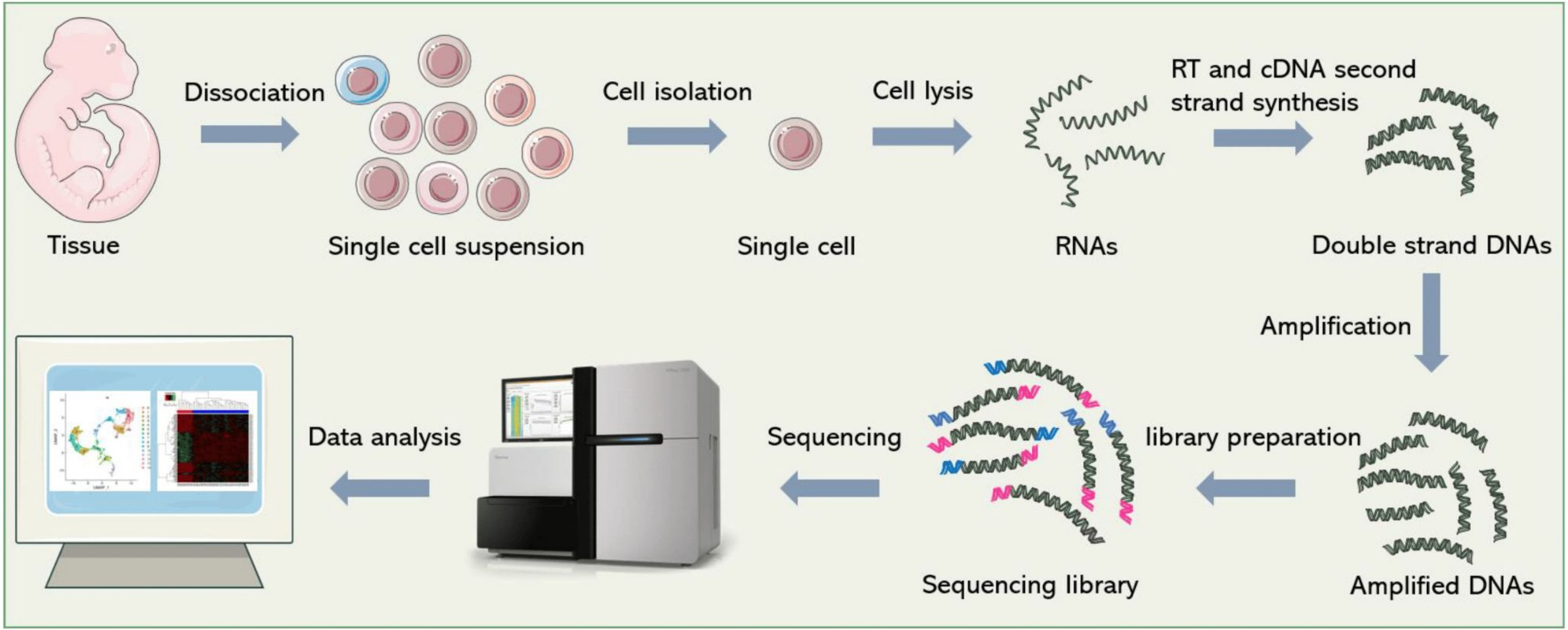
- Cancer cells
- Immune cells (T cells, macrophages, NK cells)
- Stromal cells (fibroblasts, smooth muscle cells,...)
- Endothelial cells
- Extracellular matrix (ECM), signaling molecules



Li, Z., Li, J., Bai, X. *et al.* Tumor microenvironment as a complex milieu driving cancer progression: a mini review. *Clin Transl Oncol* 27, 1943–1952 (2025).

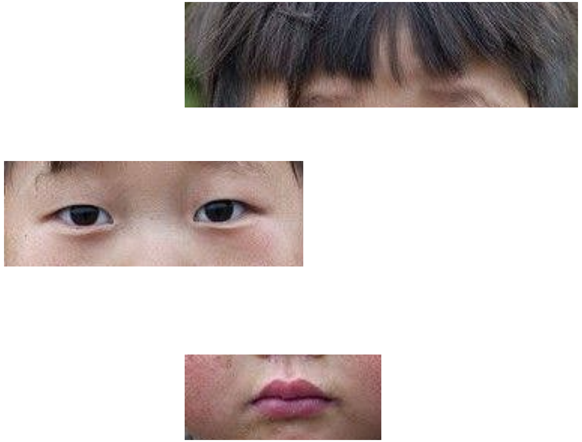
Unsupervised Learning with Single-cell Data

Single-cell technology



https://doi.org/10.3390/bios12070450

Cell signatures

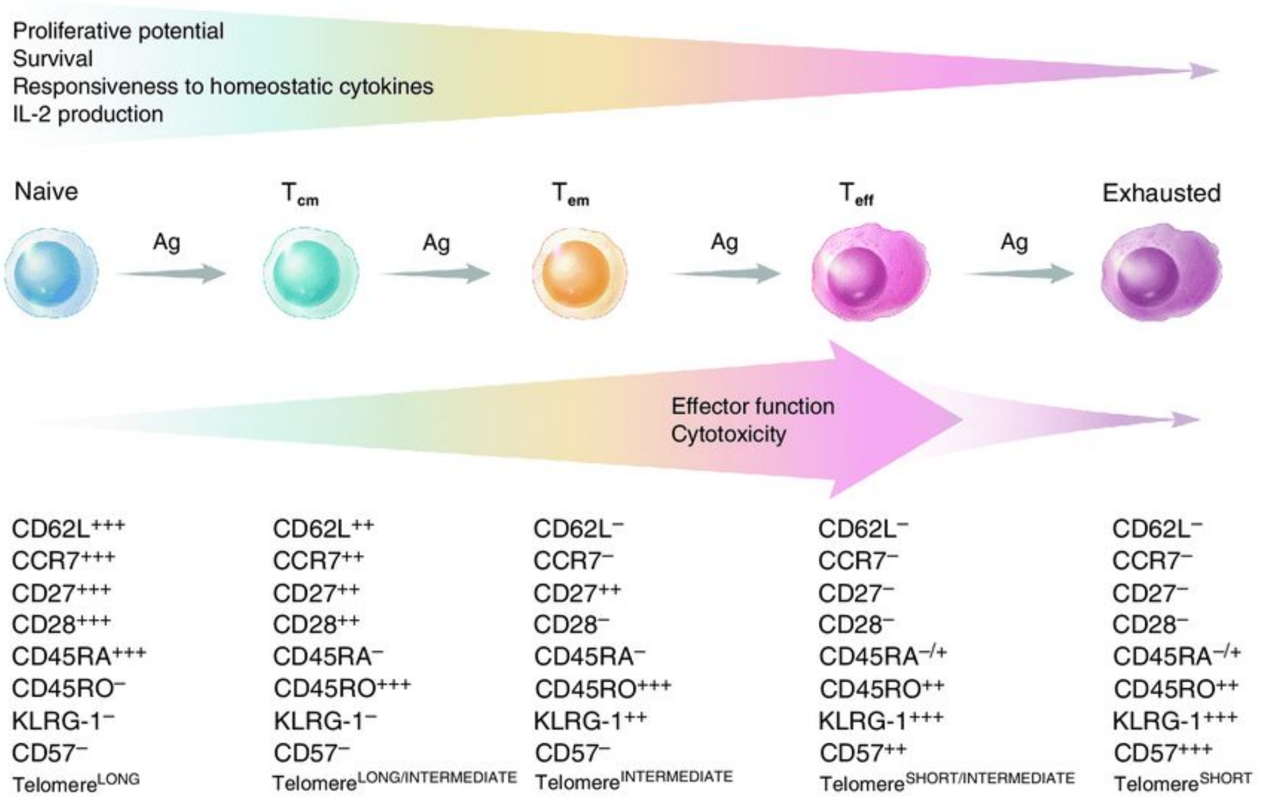


"Signature"

We differ in our characteristics...

Cell signatures

…So do the cells



“Signature”

	Cell1	Cell2	…	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
…	.	.	.	.
…	.	.	.	.
…	.	.	.	.
GeneM	25	0	.	0

https://www.researchgate.net/publication/236161870_Klebanoff_CD8_Memory_Immunol_Rev_2006/figures?lo=1

Unsupervised Learning in Single-cell Data

Question: How to regconize cell-types from a bunch of data?

Unsupervised Learning in Single-cell Data

Connecting to Biology

- Each cell type (e.g., immune cell, cancer cell, fibroblast) has a characteristic gene expression fingerprint.
- PCA captures these major axes of variation.
- Clustering uses PCA-reduced space to group similar cells.
- t-SNE/UMAP visualize the resulting embeddings.

→ “Cells with similar functional programs express similar sets of genes.

So they form clusters naturally in PCA space.”

Unsupervised Learning in Single-cell Data

Count matrix

	Cell 1	**Cell 2**	**Cell 3**	**...**	**Cell n**
Gene 1	5	0	2	...	
Gene 2	0	8	0	...	
Gene 3	3	10	1	...	
...	...	...	...	...	

- **Rows** = genes (~20,000)
- **Columns** = cells (can be 5,000–100,000+)
- **Values** = counts (how many transcript molecules from that gene were detected in that cell)

→ This is simply a high-dimensional multivariate dataset

Key properties:
High-dimensional: 20k features, 5k–100k samples
Sparse: many zeros
No labels: purely unsupervised
Noisy: biological + technical noise
Highly redundant: many genes correlated

Workflow

Gene × Cell count matrix → Normalize → Log transform → PCA → Clustering → UMAP/t-SNE

code: https://github.com/khuongduying/ds_class_sc_spatial_20251129

Single-cell data

Raw count

A data.frame: 10 × 5

	AAACCTGCACTCGACG-1	AAACCTGTCGGTCCGA-1	AAACGGGAGCCTATGT-1	AAACGGGAGGCGACAT-1	AAACGGGCACGCTTTC-1
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AL627309.1	0	0	0	0	0
AL669831.5	0	0	0	0	0
LINC00115	0	0	0	0	0
FAM41C	0	0	0	1	0
NOC2L	0	0	0	0	0
KLHL17	0	0	0	0	0
PLEKHN1	0	0	0	0	0
AL645608.8	0	0	0	0	0
HES4	0	0	0	0	0
ISG15	0	1	0	0	0

Single-cell data

Normalized & scaled count

A data.frame: 10 × 5

	AAACCTGCACTCGACG-1	AAACCTGTCGGTCCGA-1	AAACGGGAGCCTATGT-1	AAACGGGAGGCGACAT-1	AAACGGGCACGCTTTC-1
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AL627309.1	-0.03690321	-0.03690321	-0.03690321	-0.03690321	-0.03690321
AL669831.5	-0.12324993	-0.12324993	-0.12324993	-0.12324993	-0.12324993
LINC00115	-0.10663470	-0.10663470	-0.10663470	-0.10663470	-0.10663470
FAM41C	-0.19220235	-0.19220235	-0.19220235	4.35413303	-0.19220235
NOC2L	-0.30680401	-0.30680401	-0.30680401	-0.30680401	-0.30680401
KLHL17	-0.03691665	-0.03691665	-0.03691665	-0.03691665	-0.03691665
PLEKHN1	-0.08750623	-0.08750623	-0.08750623	-0.08750623	-0.08750623
AL645608.8	-0.05787025	-0.05787025	-0.05787025	-0.05787025	-0.05787025
HES4	-0.23591446	-0.23591446	-0.23591446	-0.23591446	-0.23591446
ISG15	-0.51829754	0.28697910	-0.51829754	-0.51829754	-0.51829754

Our problems & What we want

Problem

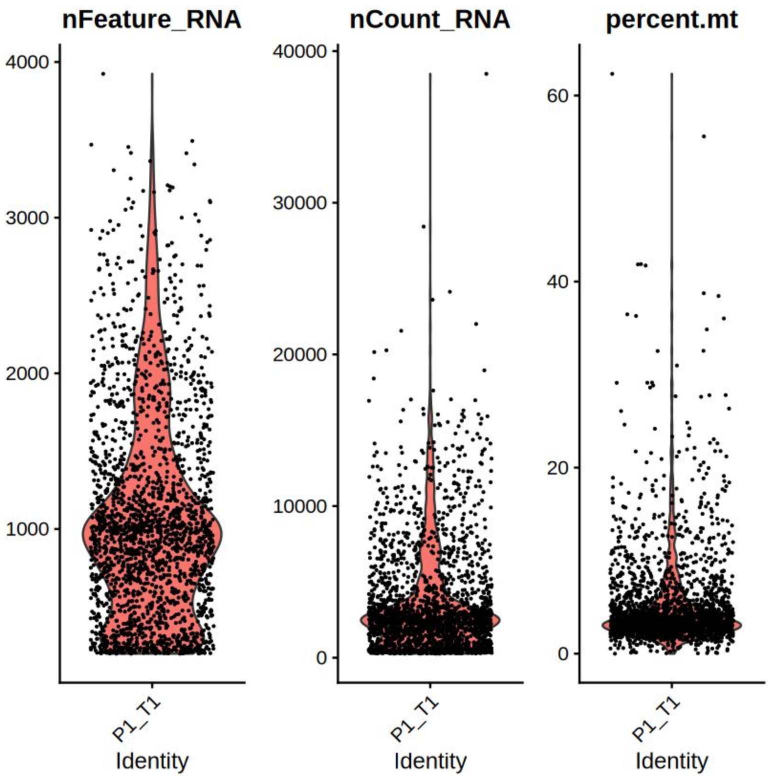
- 20,000 features (genes)
- Strong correlations (co-expressed genes)
- No labels

—

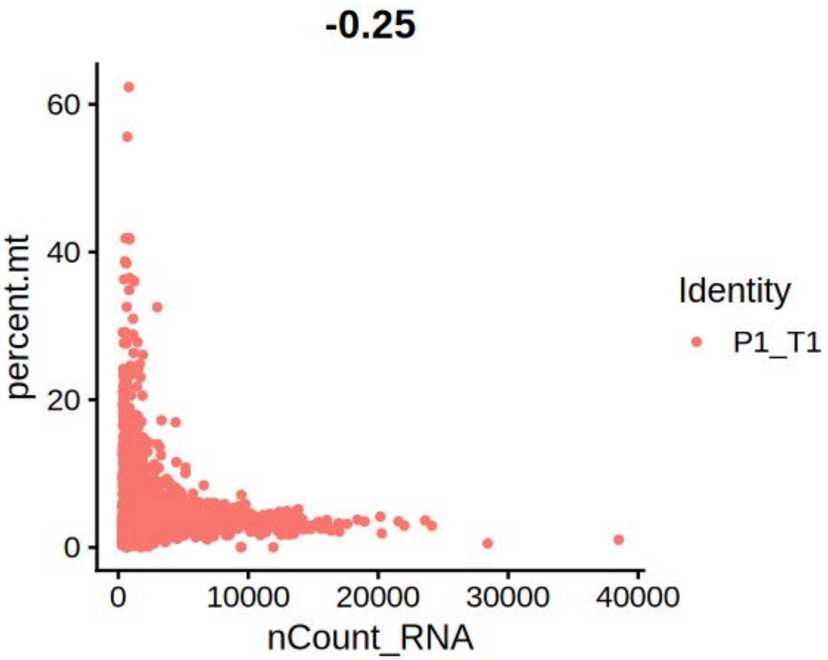
What we want:

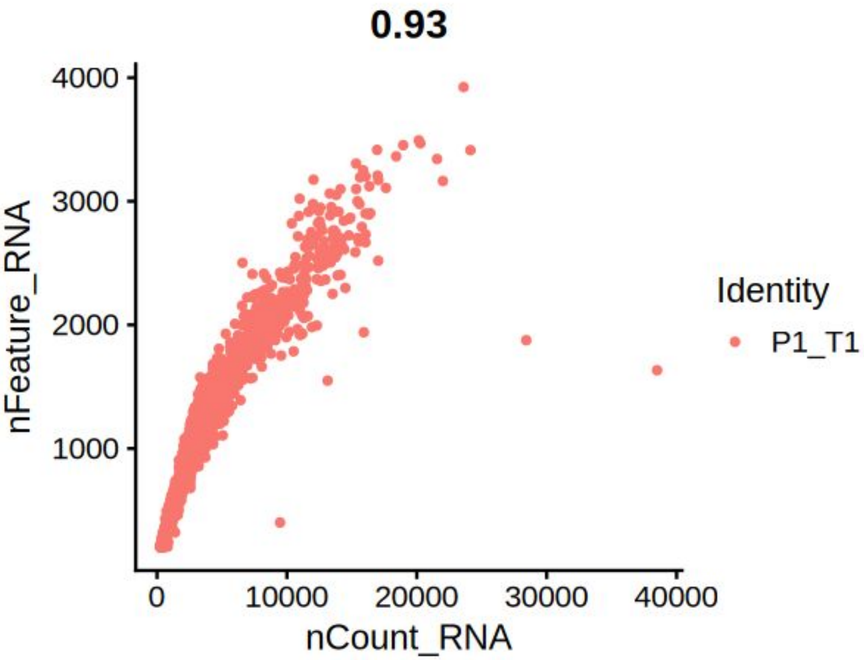
- Discover the patterns

Some pre-processing

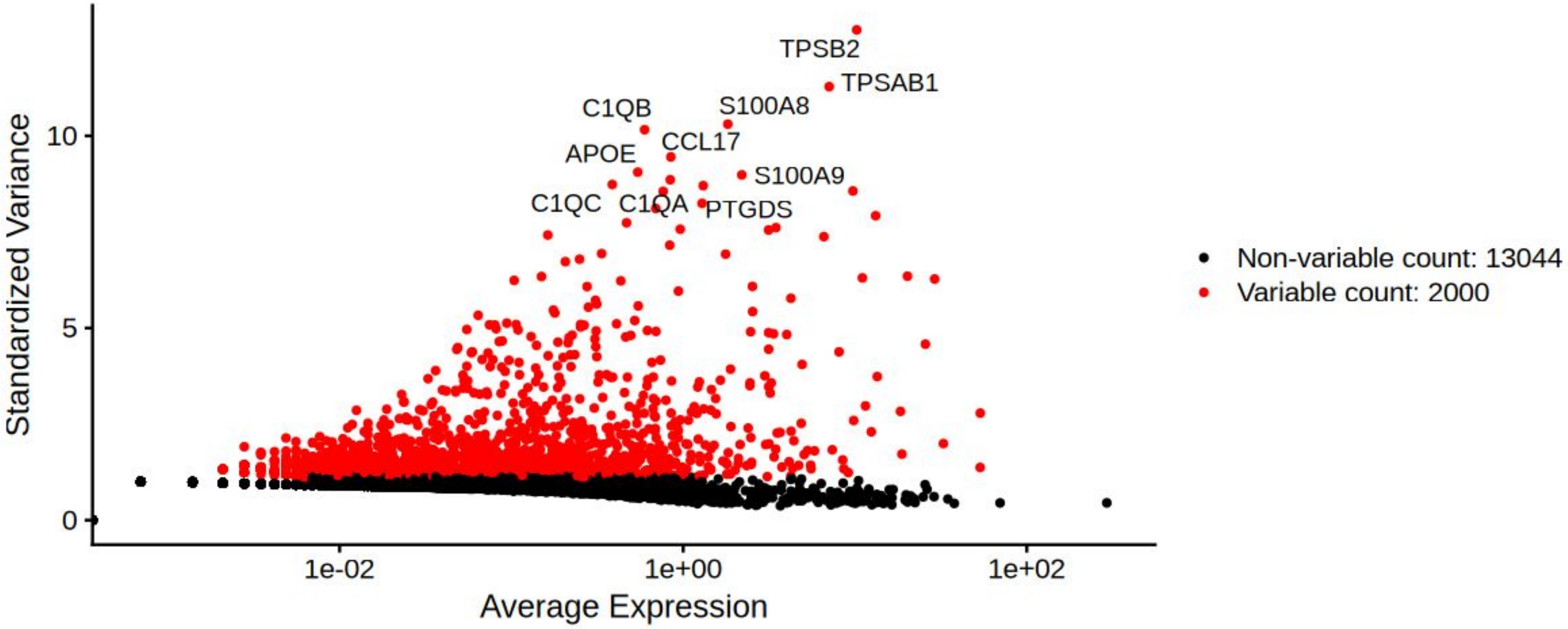


Some pre-processing





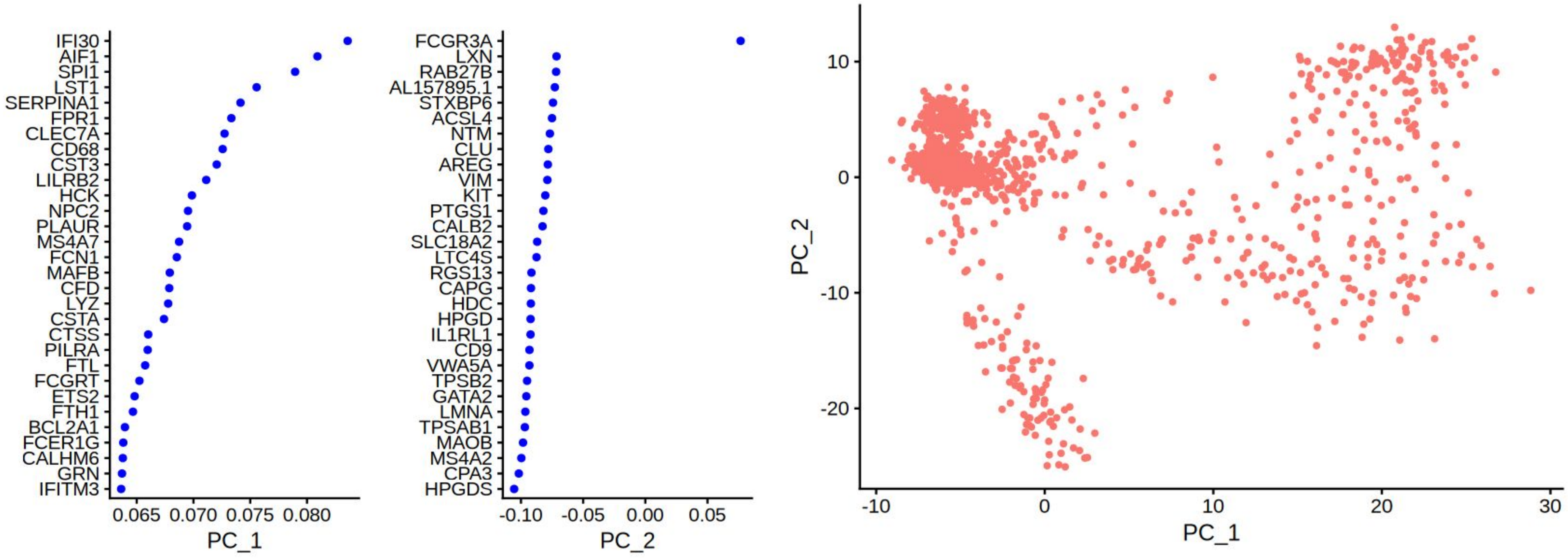
Some pre-processing



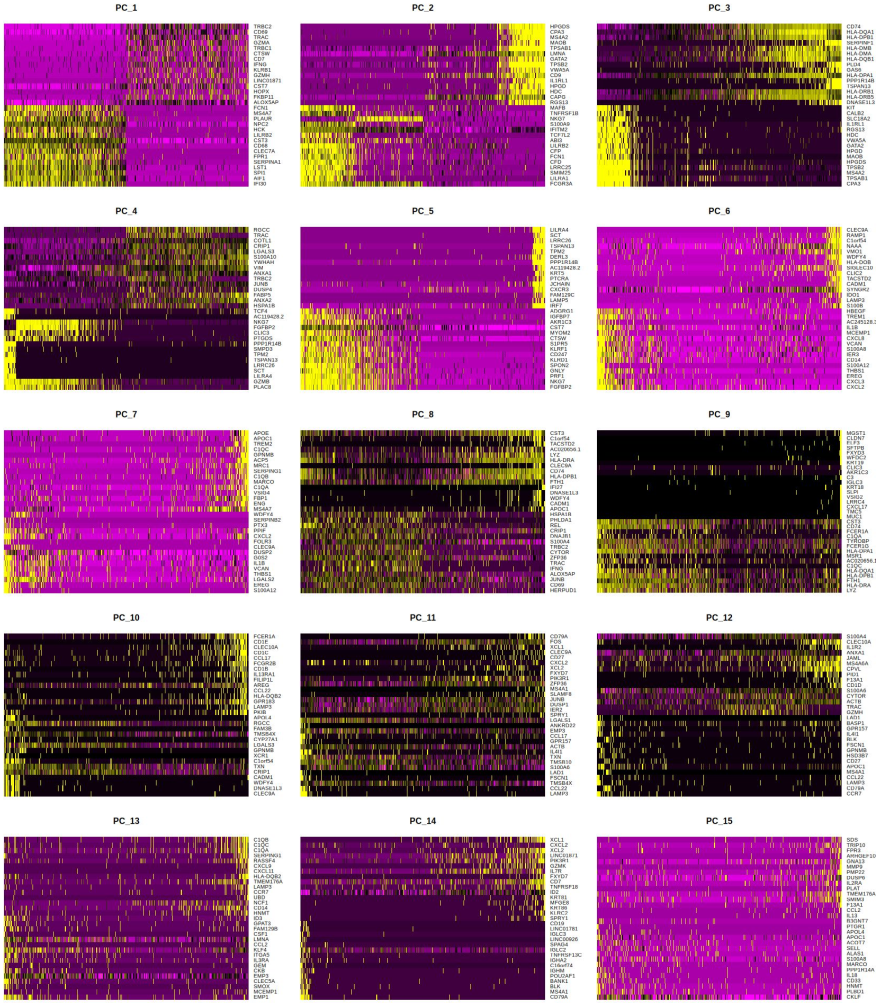
Why PCA?

- The data is extremely high-dimensional
- The data is very noisy
- Genes are highly correlated
- PCA creates meaningful distances between cells
- PCA prepares the data for visualization (UMAP/tSNE)

PCA


IFI30
AIF1
SPI1
LST1
SERPINA1
FPR1
CLEC7A
CD68
CST3
LILRB2
HCK
NPC2
PLAUR
MS4A7
FCN1
MAFB
CFD
LYZ
CSTA
CTSS
PILRA
FTL
FCGRT
ETS2
FTH1
BCL2A1
FCER1G
CALHM6
GRN
IFITM3
0.065 0.070 0.075 0.080
PC_1
FCGR3A
LXN
RAB27B
AL157895.1
STXBP6
ACSL4
NTM
CLU
AREG
VIM
KIT
PTGS1
CALB2
SLC18A2
LTC4S
RGS13
CAPG
HDC
HPGD
IL1RL1
CD9
VWA5A
TPSB2
GATA2
LMNA
TPSAB1
MAOB
MS4A2
CPA3
HPGDS
-0.10 -0.05 0.00 0.05
PC_2
10
0
-10
-20
PC_2
-10
0
10
20
30
PC_1

PCA


PC_1
PC_2
PC_3
PC_4
PC_5
PC_6
PC_7
PC_8
PC_9
PC_10
PC_11
PC_12
PC_13
PC_14
PC_15

Clustering

Goal: group cells with similar gene-expression patterns → cell types or functional states.

Once we have PCA-reduced coordinates (e.g., 30–50 PCs), each cell becomes a point in a lower-dimensional space.

→ This creates the perfect setup for unsupervised learning.

Why clustering is needed?

Remember: we have no labels.

We do not know which cells are cancer cells, immune cells, fibroblasts, etc.

But cells with similar gene-expression programs naturally form groups.

So clustering is essentially:

“Discovering structure in a cloud of high-dimensional biological points.”

How clustering is done

Step 1 — Distance in PCA space

We compute distances between cells in the PCA-reduced space:

- Euclidean
- or more commonly **k-nearest neighbors (kNN)** graph

This creates a *graph structure*:

- each cell = node
- edges = similar expression profiles

This is very standard for large, high-dimensional data.

How clustering is done

Step 2 — Graph-based clustering

Modern single-cell tools (Seurat, Scanpy) mostly use:

Leiden or Louvain clustering

- Input: kNN graph
- Output: communities of cells
- These communities are the “clusters” we visualize

“We perform community detection on a graph where cells are connected to similar cells.”

How to visualize clusters?

Once we have clusters in PCA space, we project them into:

- **t-SNE**, or
- **UMAP**

These methods don't cluster — they *visualize*.

> **"Clustering happens in PCA space.**
> **UMAP/t-SNE are just for illustration."**

Minimal biology interpretation (simple language)

Once clusters are found, the biological step is:

- identify **marker genes** enriched in each cluster
- name the clusters (immune cells, cancer cells, fibroblasts, etc.)

“Clusters reflect groups of cells performing similar functions — often different cell types or cancer subpopulations.”

Workflow

Gene × Cell matrix

↓ normalization + log

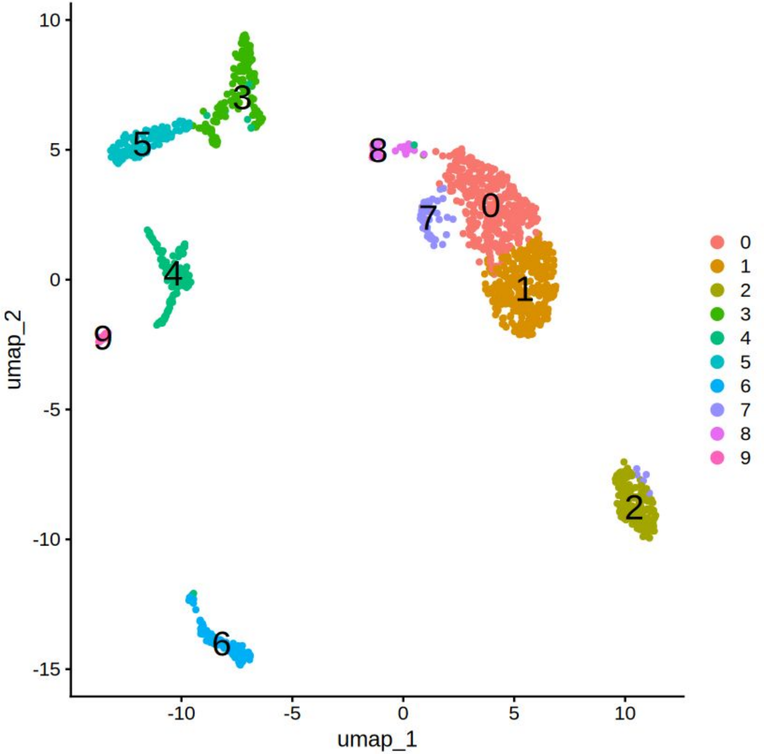
↓ PCA (reduce to ~30 PCs)

↓ kNN graph (similar cells)

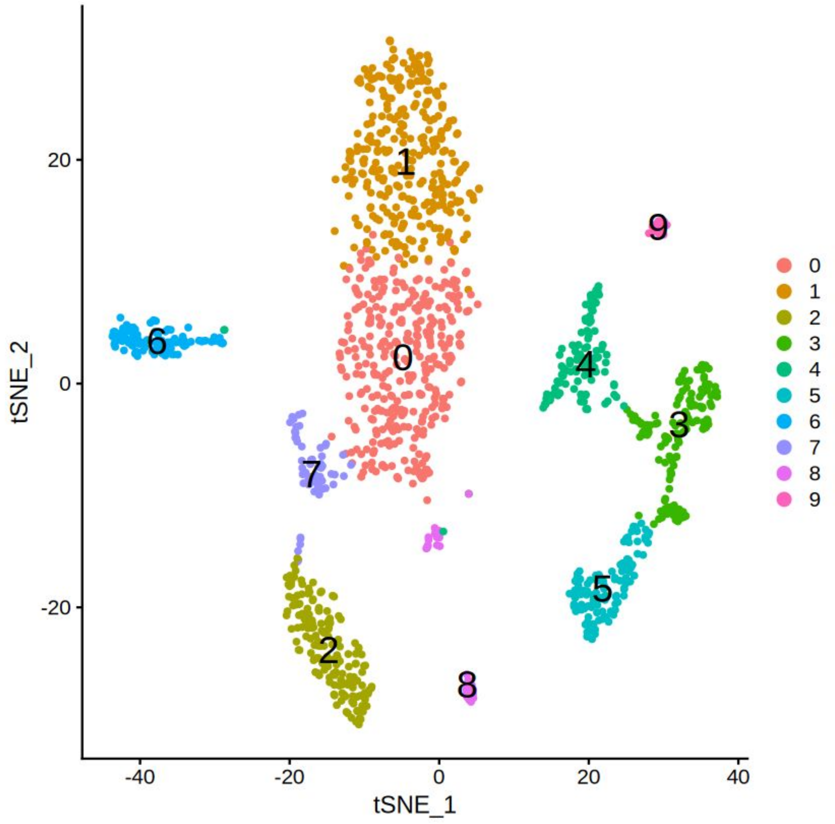
↓ Leiden/Louvain clustering (community detection)

↓ Visualize with UMAP/tSNE

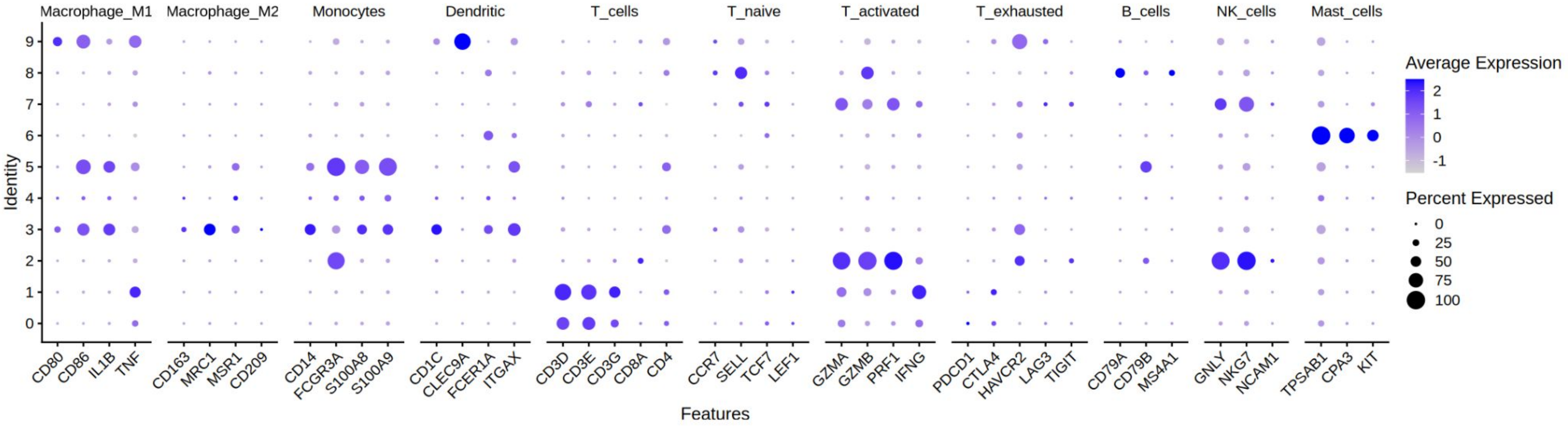
UMAP (Uniform Manifold Approximation and Projection for Dimension Reduction)



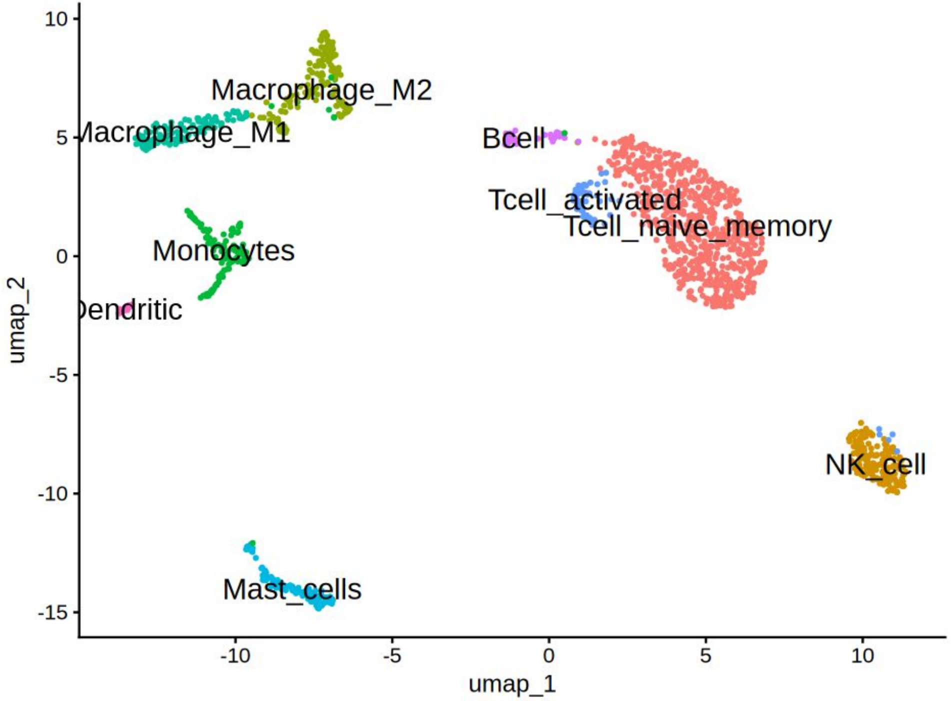
tSNE (t-distributed Stochastic Neighbor Embedding)



Cell-type Markers



Cell-type Annotation



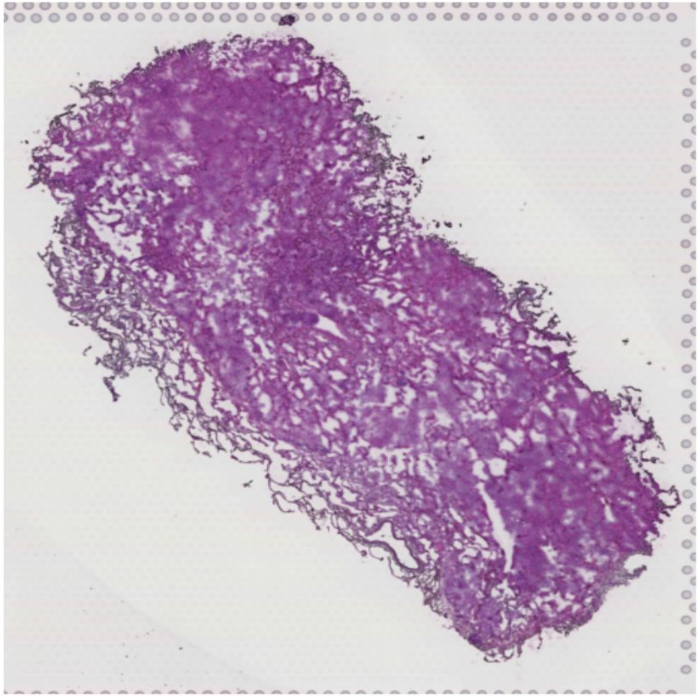
Spatial Autocorrelation in Spatial Transcriptomics

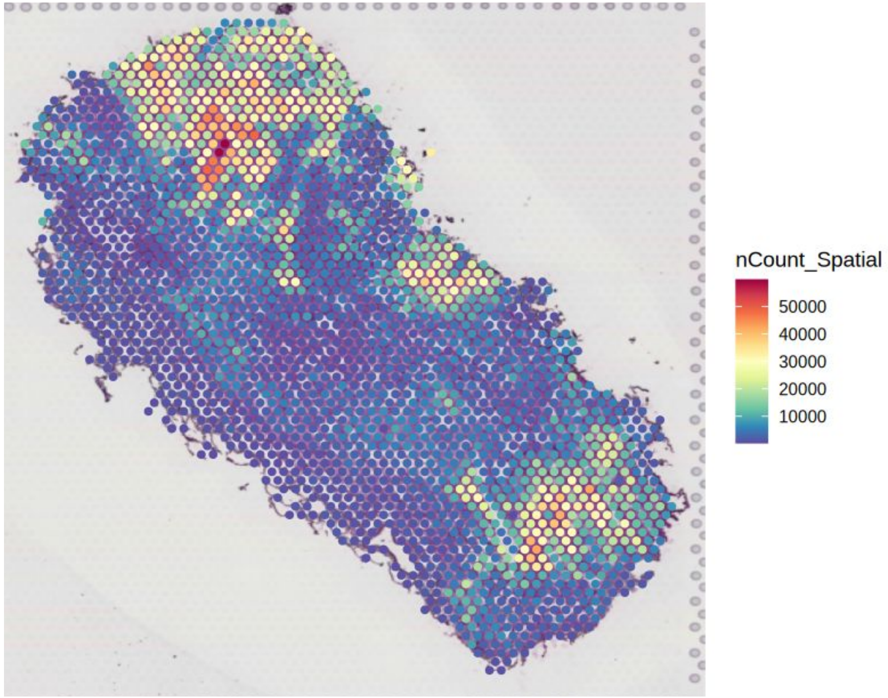
Spatial Autocorrelation in ST (Moran's I)

Moran's I = spatial correlation of gene expression

Question: "Are similar values near each other? Or randomly scattered?"

Visium Spatial Transcriptomics

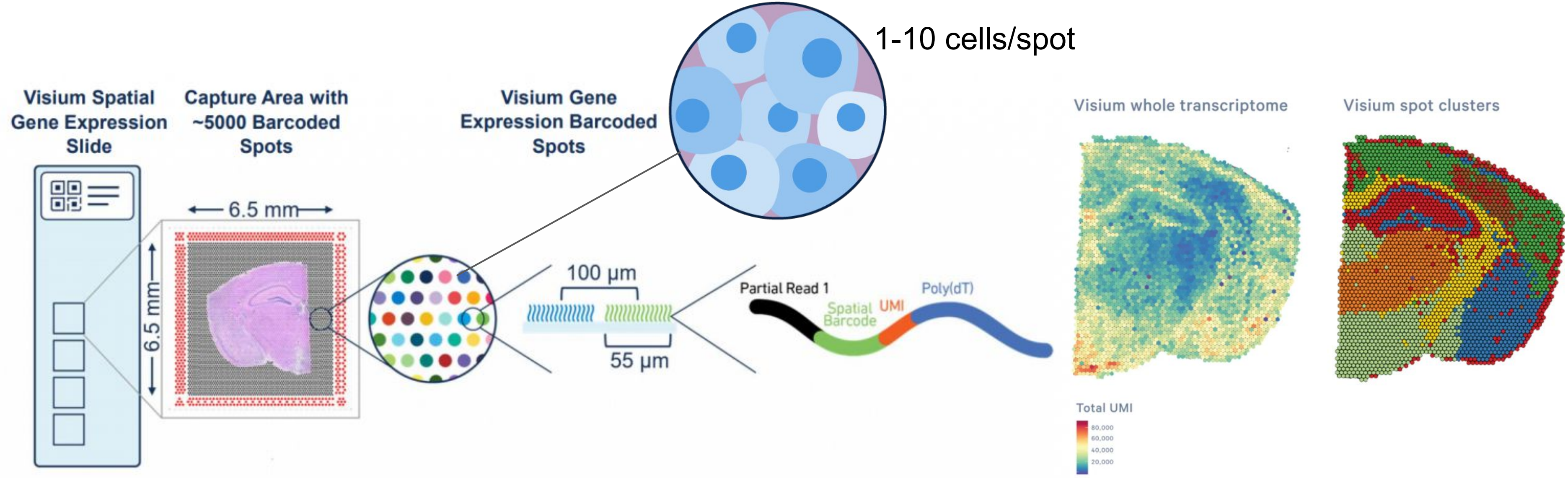




From a lung cancer tissue section…

…to a lung cancer tissue section with thousands of gene expression information

Visium Spatial Transcriptomics

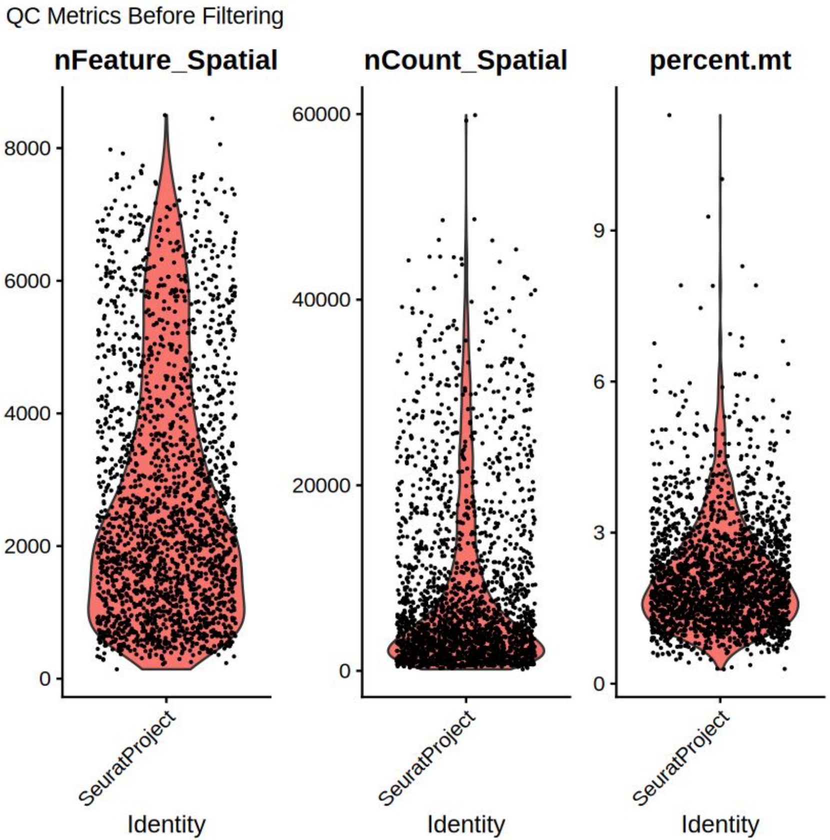


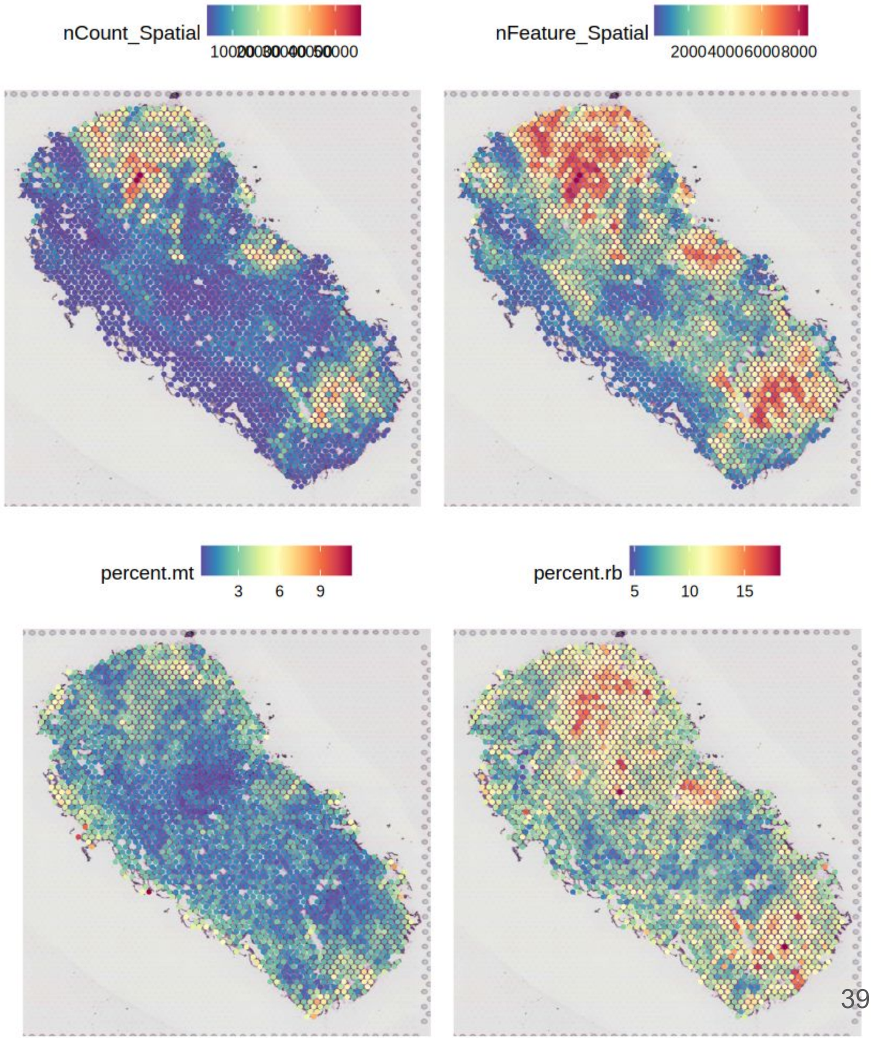
High throughput: Capture 15,000-18,000 genes

➤ High throughput - Comparable to single-cell

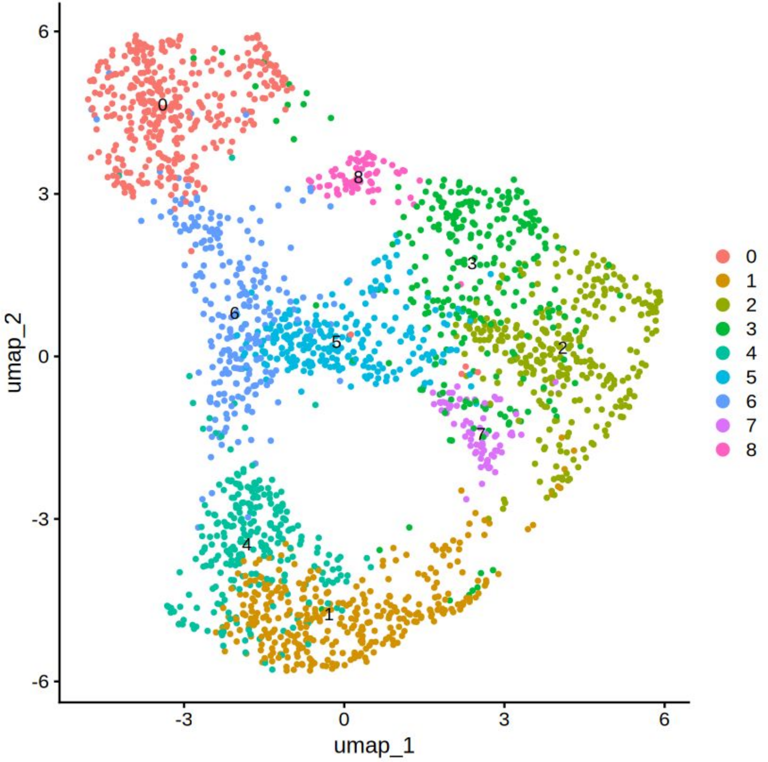
https://www.10xgenomics.com/analysis-guides/integrating-single-cell-and-visium-spatial-gene-expression-data
https://www.10xgenomics.com/support/spatial-gene-expression-fresh-frozen

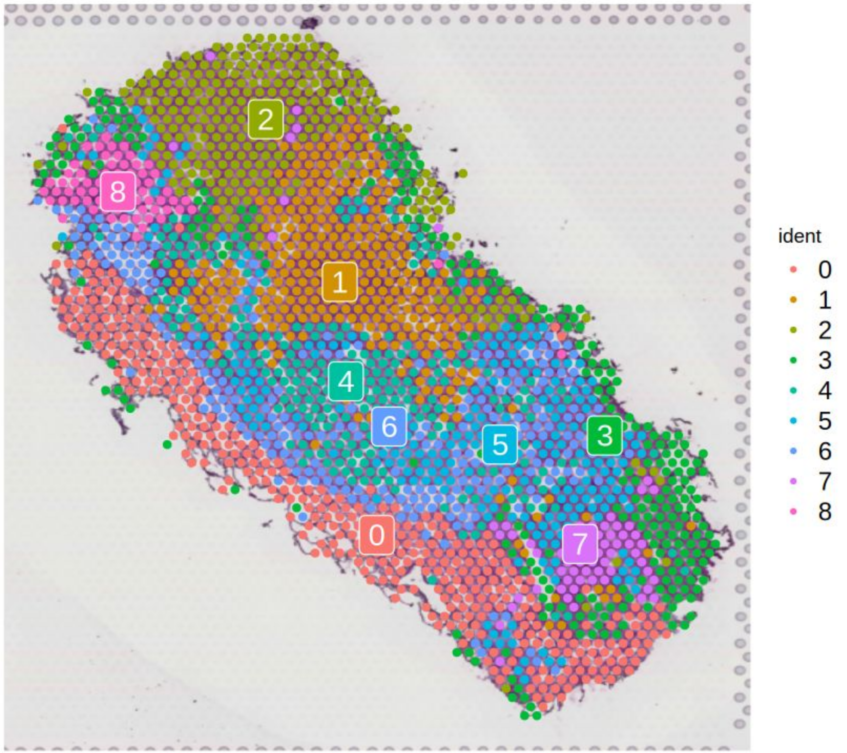
Explore QC metrics





Dimensional Reduction & Clustering



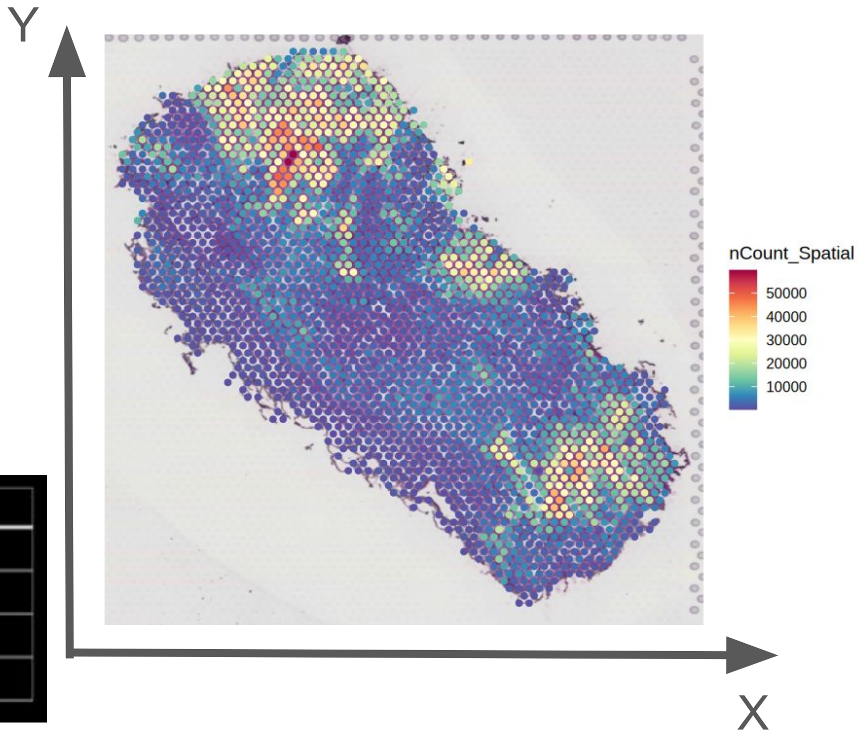


Spatial Transcriptomics data

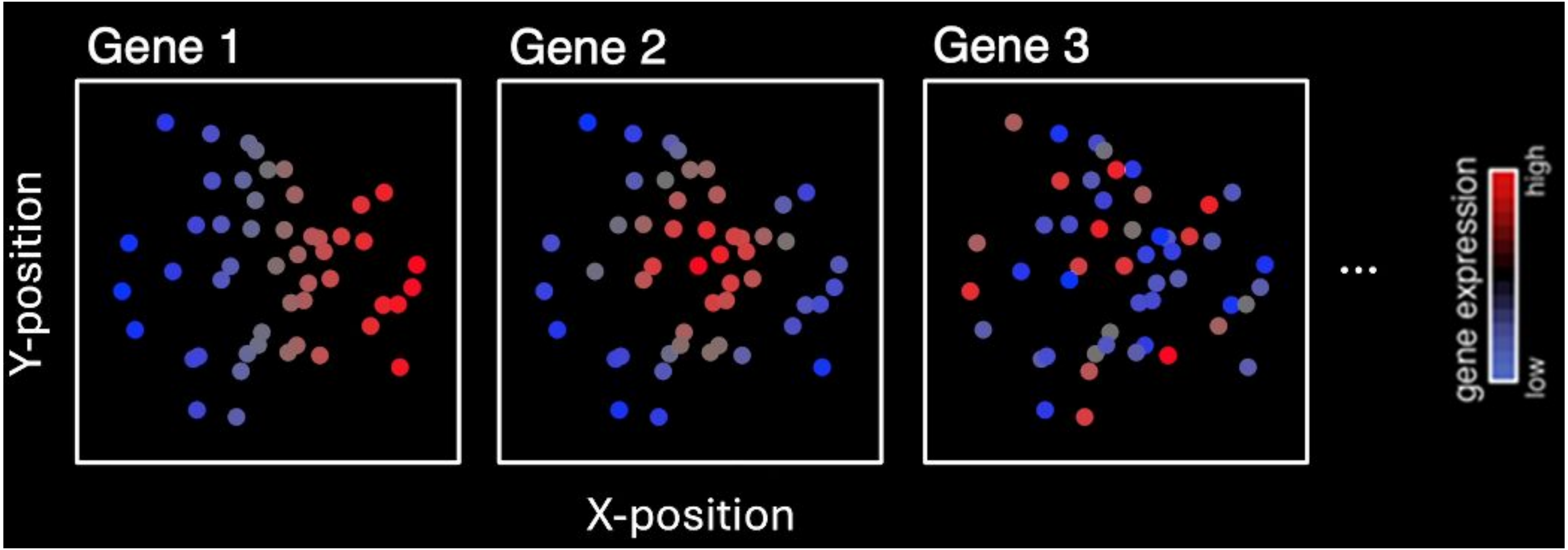
In Spatial Transcriptomics:

- each “spot” has a coordinate (x, y)
- each spot has a gene expression value
- we want to know **whether the gene expression pattern is random, clustered, or dispersed in space**

	Spatial X	Spatial Y	Gene 1	Gene 2	Gene 3	...
Cell1	1.5	43.5	23	86	153	...
Cell2	24.4	12.4	356	45	56	...
Cell3	57.4	75.3	387	23	34	...
...	...	...	...	...	...	...



Spatially Variable Features: Moran's I

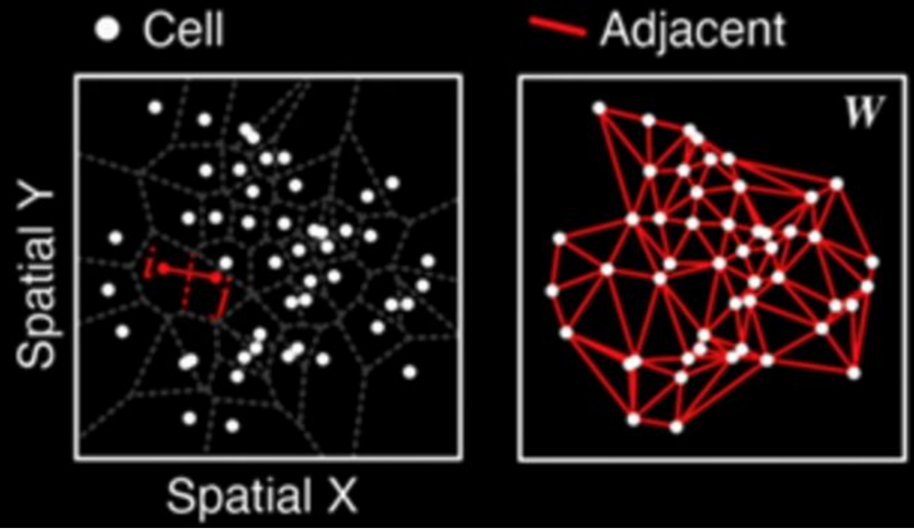


We don't want to look for pattern manually → use statistics to filter and rank

Moran's I

$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N \boxed{W_{ij}}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i^N (x_i - \bar{x})^2}$$

$$W_{ij} = \begin{cases} 1 & if\ cell_i\ and\ cell_j\ are\ adjacent \\ 0 & if\ cell_i\ and\ cell_j\ are\ not\ adjacent \end{cases}$$



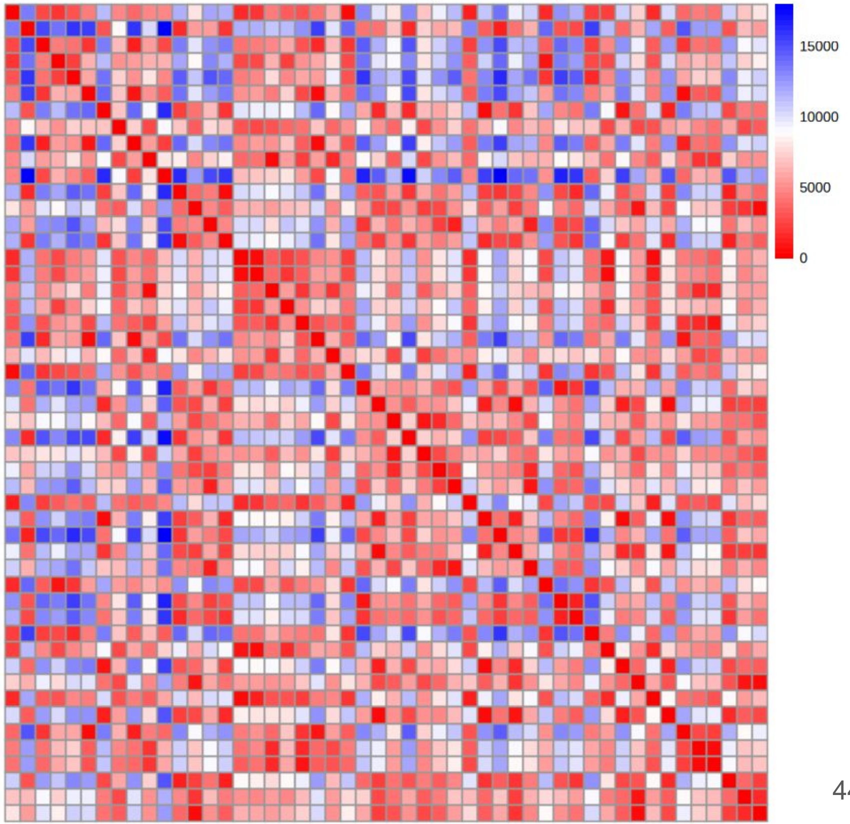
Constructing a distance matrix

2D spatial coordinates (X,Y)

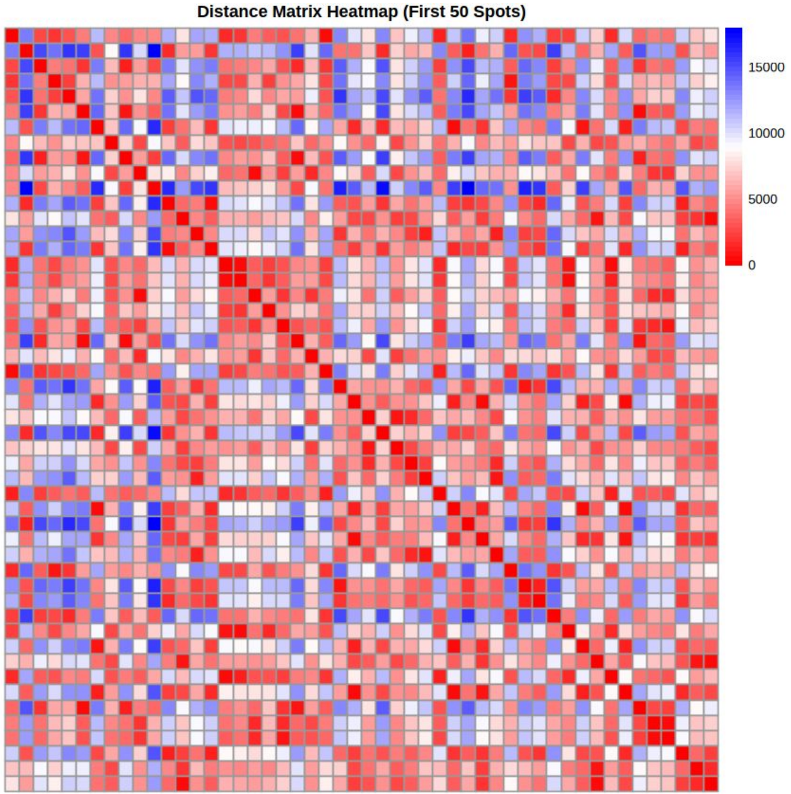
	x	y
	<dbl>	<dbl>
AAACAAGTATCTCCCA-1	10773	12677
AAACAATCTACTAGCA-1	1794	6199
AAACATTTCCCGGATT-1	12875	12128
AAACCCGAACGAAATC-1	9818	14104
AAACCGGAAATGTTAA-1	11538	15093
AAACCTAAGCAGCCGG-1	13639	10591
AAACGAAGAACATACC-1	2367	8505
AAACGAGACGGTTGAT-1	7908	10152
AAACGGGCGTACGGGT-1	13639	11469
AAACTGCTGGCTCCAA-1	9818	8834

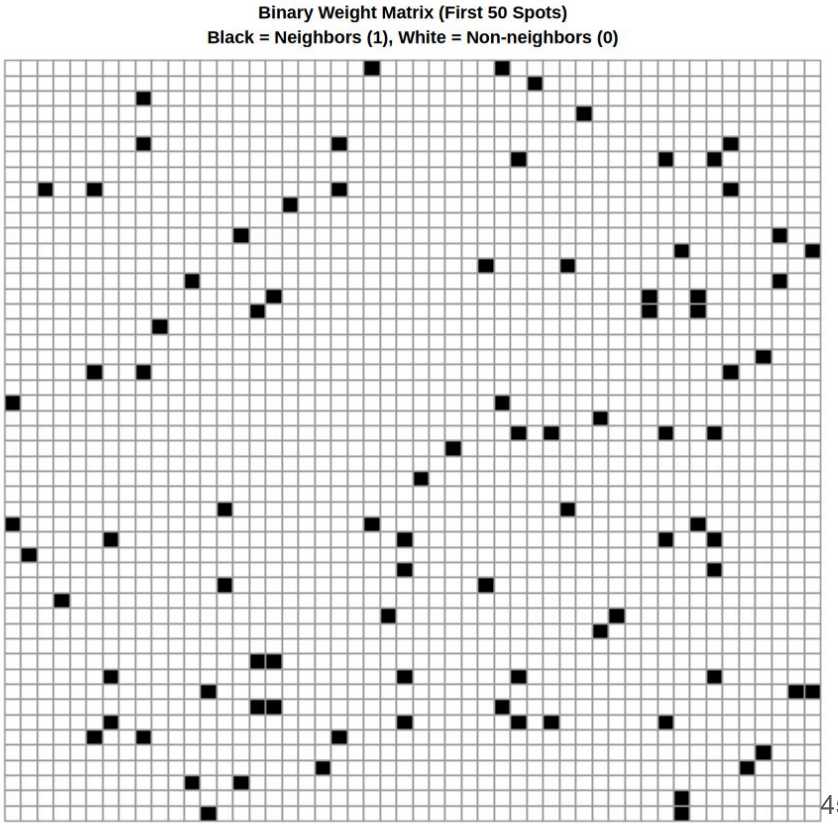
	AAACAAGTATCTCCCA-1	AAACAATCTACTAGCA-1	AAACATTTCCCGGATT-1
AAACAAGTATCTCCCA-1	0.00	13560.25	2660.77
AAACAATCTACTAGCA-1	13560.25	0.00	15391.96
AAACATTTCCCGGATT-1	2660.77	15391.96	0.00
AAACCCGAACGAAATC-1	2102.98	13795.30	4458.11
AAACCGGAAATGTTAA-1	3103.78	16157.76	3983.49
AAACCTAAGCAGCCGG-1	4341.43	15472.25	2102.16
AAACGAAGAACATACC-1	11493.46	2910.15	13613.09
AAACGAGACGGTTGAT-1	4677.15	8916.88	6547.02
AAACGGGCGTACGGGT-1	3809.18	15878.14	1235.70
AAACTGCTGGCTCCAA-1	4849.85	10343.68	5503.96

Distance Matrix Heatmap (First 50 Spots)



From distance matrix to weight matrix (Wij)

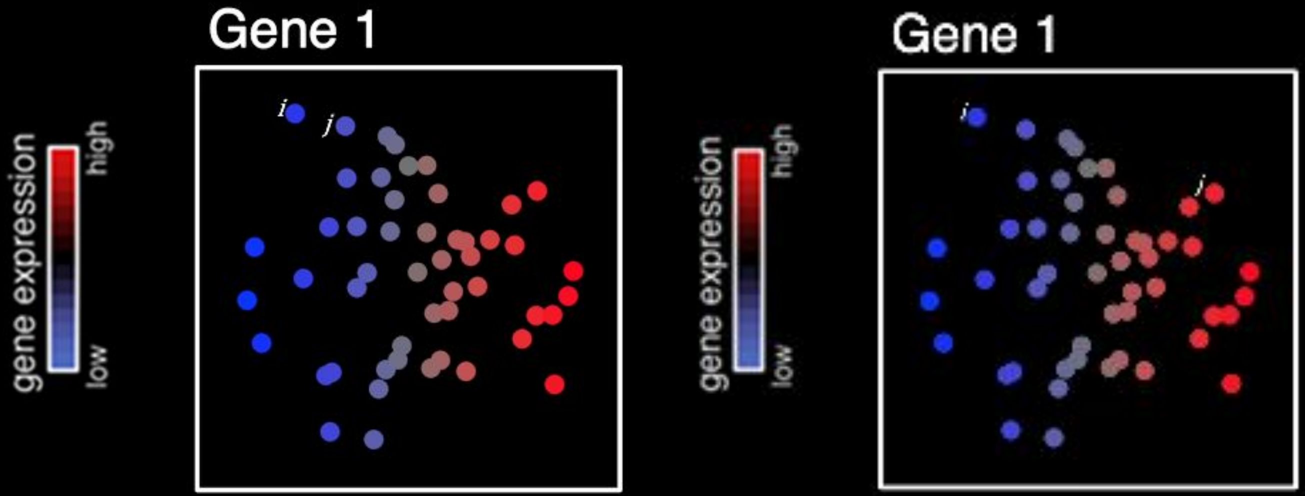




Moran's I

$$\boxed{I} = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_i^N (x_i - \overline{x})^2}$$

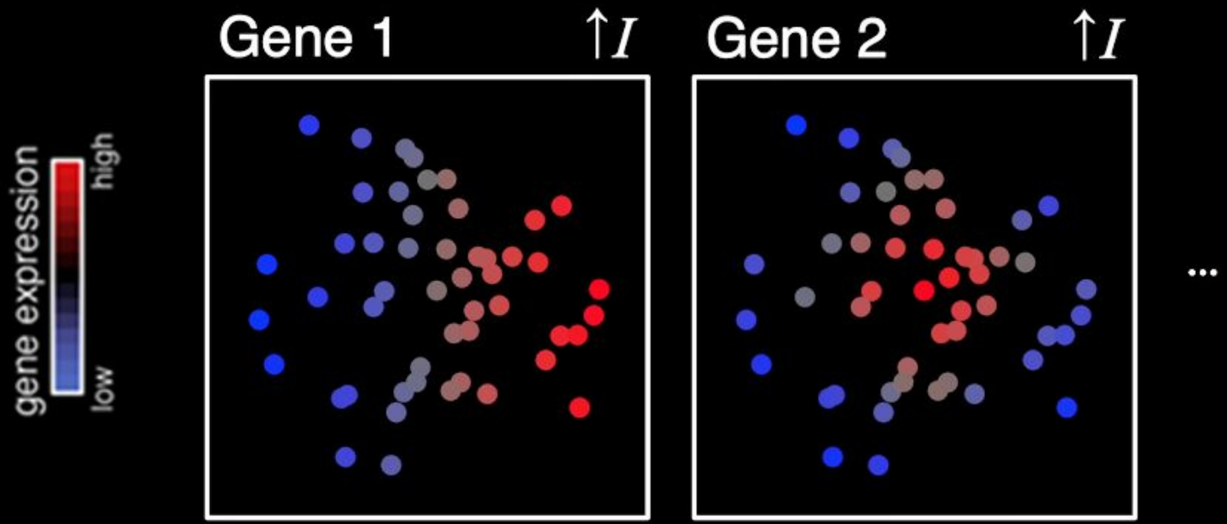
x_i = expression of a particular gene in $cell_i$
x_j = expression of a particular gene in $cell_j$



Moran's I

$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_i^N (x_i - \overline{x})^2}$$

x_i = expression of a particular gene in cell$_i$
x_j = expression of a particular gene in cell$_j$



Moran's I

$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i^N (x_i - \bar{x})^2}$$

x_i = expression of a particular gene in $cell_i$
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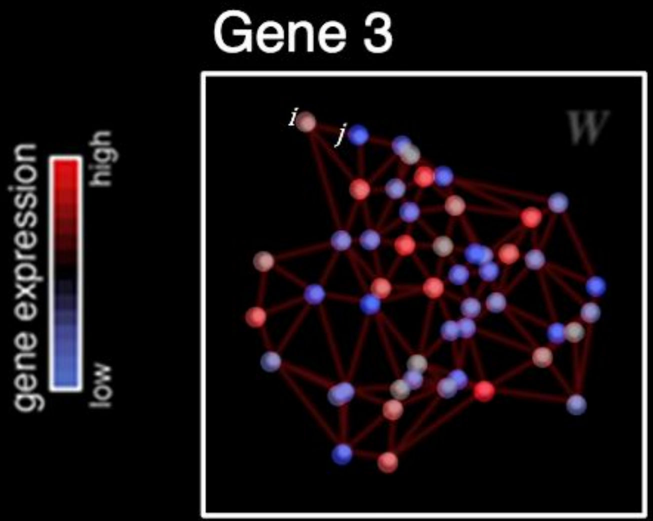
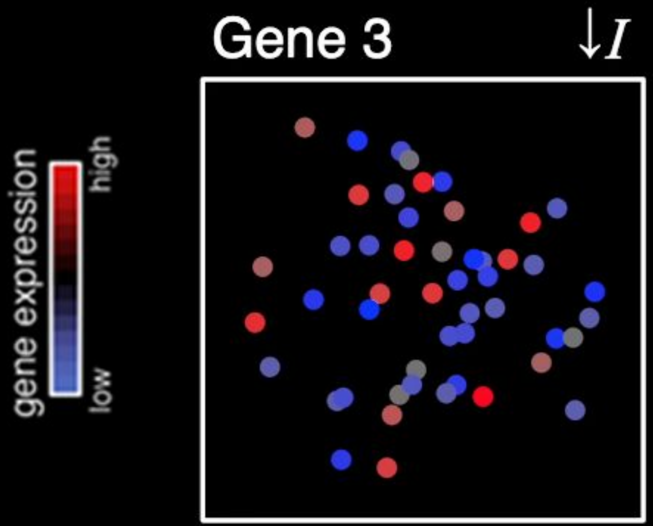


Moran's I

$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_i^N (x_i - \overline{x})^2}$$

x_i = *expression of a particular gene in* $cell_i$
x_j = *expression of a particular gene in* $cell_j$



Moran's I

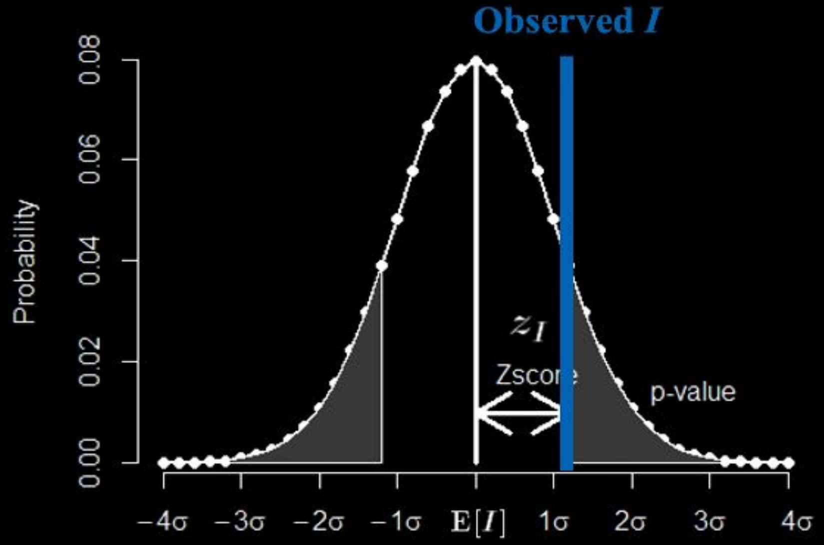
$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_i^N (x_i - \overline{x})^2}$$

H_0 = gene is not spatially variable
H_A = gene is spatially variable

$$z_I = \frac{I - \mathrm{E}[I]}{\sqrt{\mathrm{V}[I]}}$$

<u>under spatial randomness</u>

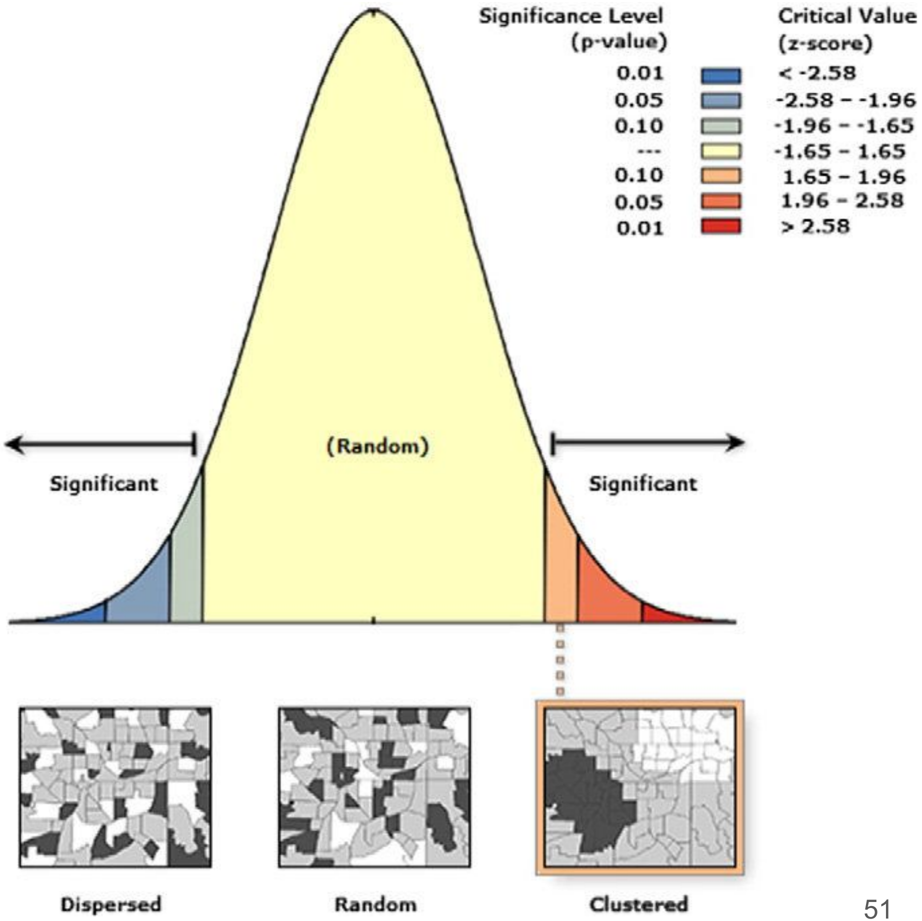
$$\mathrm{E}[I] = -1/(N-1)$$
$$\mathrm{V}[I] = \mathrm{E}[I^2] - \mathrm{E}[I]^2$$



Moran’s I Statistics

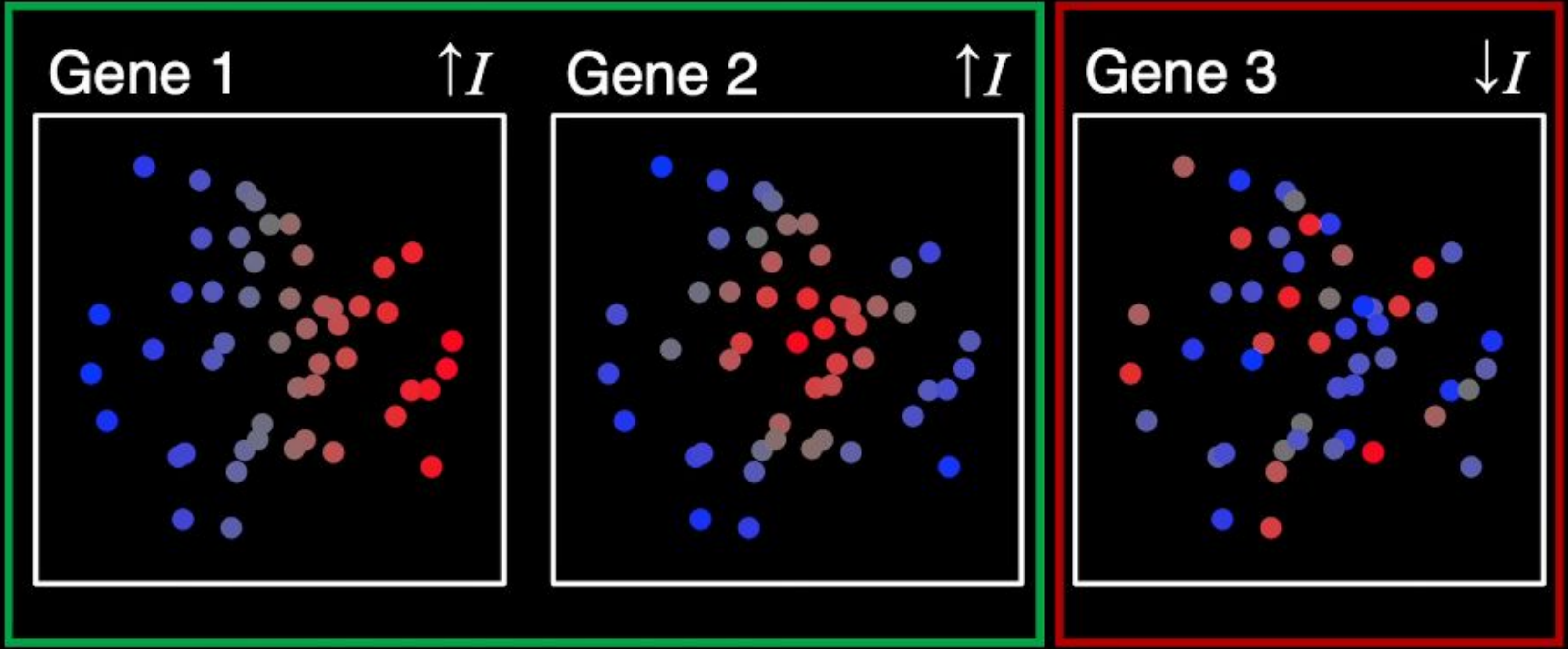
Moran’s I

$$I = \frac{N}{\sum_i^N \sum_j^N W_{ij}} \frac{\sum_i^N \sum_j^N W_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_i^N (x_i - \overline{x})^2}$$



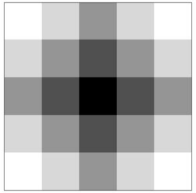
$H_0 =$ *gene is not spatially variable*

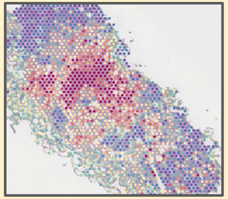
$H_A =$ *gene is spatially variable*



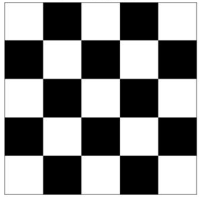
Spatially variable features (SVGs)

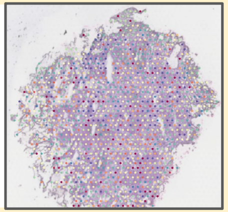
High Moran's I (Clustered)





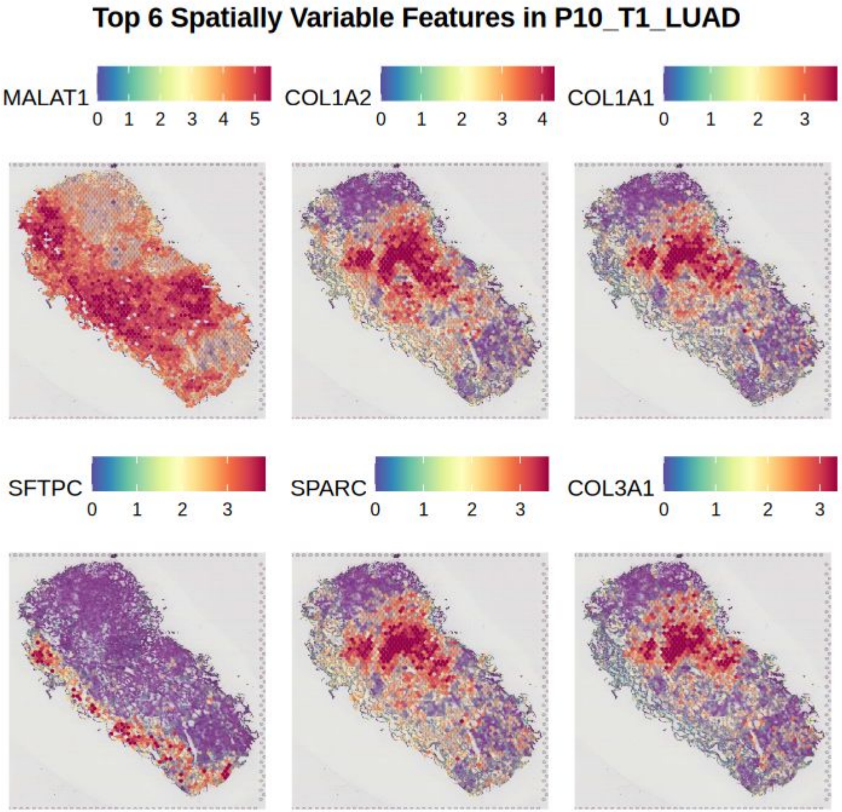
Low Moran's I (Random/Dispered)



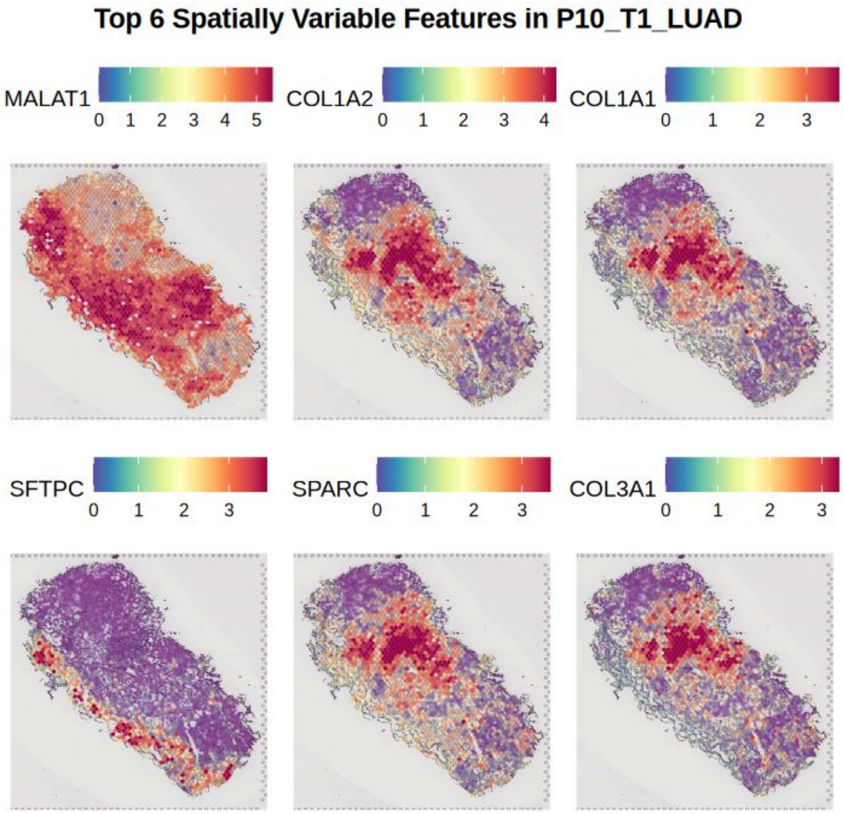


A data.frame: 10 × 3

	MoransI_observed	MoransI_p.value	moransi.spatially.variable.rank
	<dbl>	<dbl>	<int>
AC007952.4	0.5102738	0	1
HLA-DRB1	0.5081747	0	2
COL1A2	0.5047799	0	3
SFTPC	0.4960138	0	4
MALAT1	0.4930120	0	5
COL1A1	0.4831377	0	6
SFTPB	0.4735983	0	7
SPARC	0.4707774	0	8
NAPSA	0.4661206	0	9
HLA-DPB1	0.4647361	0	10



Biological Interpretation



- Collagen expression → fibroblast niche/fibrosis region → Cancer progression

- Surfactant (SFTPC) → Normal lung cells

→ Cancer cells alter the microenvironment of the tissue to support their progression and invasion.

Conclusion

1. **Unsupervised learning** finds structure in single-cell data by grouping cells based on shared expression patterns without needing labels.
2. **Spatial autocorrelation** reveals whether genes or cell states form meaningful spatial patterns rather than occurring randomly across tissue.

Take-home message:

Let the data speak — both in how cells cluster and where they live.

Reference

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